

Shrey Agarwal

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EDUCATION

Georgia Institute of Technology, Atlanta, GA

May 2027

Bachelor of Science in Computer Science, Minor in Financial Technology

GPA: 4.0/4.0

Concentrations: Distributed Systems and Embedded Devices

Relevant Coursework: Data Structures & Algorithms, Discrete Math, Database Systems, Programming for HW/SW Systems, Digital System Design, Computer Architecture, Computer Networking, Intro to Artificial Intelligence, Robotics and Perception

SKILLS & INTERESTS

Proficient: Java, Python, SQL, JavaScript, React.js, C++, Linux, Github, AWS, Docker, PyTorch, MongoDB, Agile Framework

Interests: Software Engineering, Machine Learning & AI, Autonomous Robotics, Computer Vision, Financial Markets

RELEVANT EXPERIENCE

Software Engineering Intern, *CarMax*

May 2025 — August 2025

- Thinned bonus sync workflow by 99.5% by parameterizing Azure Data Factory ETL pipelines for 500+ car dealerships
- Implemented Azure Functions to validate 2500+ quarterly incentive samples for Snowflake to SAP transformations
- Reduced cloud compute costs by 35% via architecting optimized Medallion staging for SAP bonus sync ETL pipelines

Discrete Math Teaching Assistant, *Georgia Institute of Technology*

January 2025 — Current

- Teach weekly recitation of 20+ students and mentor 500+ students one-on-one during office hours on lectures/homework
- Grade proof-based written problem sets and 6 semesterly exams in logic, set theory, combinatorics, and graph theory

Machine Learning Research Assistant, *Georgia Institute of Technology*

August 2024 — April 2025

- Developed SLAM algorithm for novel fire suppression system to detect and mitigate fires from an autonomous drone
- Analyzed, transformed, and cleaned data from 5 sensors, including camera, LIDAR, depth sensor, and heat sensor
- Trained and validated multi-modal machine learning fire detection algorithm with 95% precision and 92% recall

Software Development Intern, *LCKR Startup*

May 2023 — August 2023

- Designed and deployed full-stack peer-to-peer rental marketplace via Sharetribe Flex, TypeScript, HTML, Tailwind CSS
- Facilitated 10,000+ exchanges between 50+ lenders and 100+ renters, supporting user transactions via Stripe Payments API
- Scaled platform by 150% within 3 months by establishing 10+ partnerships with on-campus facilities and local businesses

PROJECTS | agshrey.dev

VibeCheck, AI Code Analysis Dev Tool | <https://github.com/agshrey/vibecheck-extension>

April 2025 — August 2025

- Developed VS Code extension that blocks inserting copy-pasted and AI-generated code, cutting debugging time by 40%
- Integrated React dashboard to provide 9+ metrics on vibe coding habits and generate custom learning paths for skill-building
- Leveraged SBERT model to vectorize code snippets in 384 dimensions to determine semantic similarities in vibe code history
- Tech Stack: TypeScript, Hugging Face, SBERT, Gemini API, VS Code Extension API, Supabase, Tailwind CSS, HTML

PerfectPunch, AI Boxing Trainer | <https://devpost.com/software/perfectpunch>

February 2025

- Implemented AI-powered boxing trainer using computer vision and pose estimation to classify punches with 98% accuracy
- Transformed human wireframe keypoints into 2D vectors, generating 5,000+ plotted samples for further motion analysis
- Tech Stack: Python, Django, JavaScript, React.js, Numpy, Pandas, Matplotlib, OpenCV, MediaPipe, Taipy, Gemini API

SafeWay, Safety-Optimized Navigation App | <https://devpost.com/software/saferoute-2iyzml>

September 2024

- Engineered navigation app that ranks three custom routes based on priority for crime safety, lighting, walkability, road quality
- Scraped and erased 500k+ data points, integrated optimized querying, LLM search, Docker and Terraform deployment
- Tech Stack: Python, Flask, JavaScript, MongoDB Atlas, Pymongo, Docker, Terraform

ACTIVITIES

Lead Backend Developer, *iOS Club @ Georgia Institute of Technology*

August 2024 — December 2024

- Led team of 5 developers to build database and authentication system for fitness social media app using Firebase
- Integrated backend authentication flows with frontend using Swift, enabling seamless user login and profile management